

The periodic table

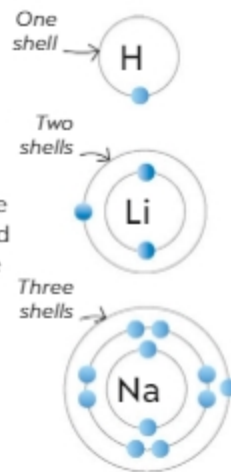
Many scientists had tried to arrange the known elements into an organized list long before Russian chemist Dmitri Mendeleev produced his first table of elements in 1869. Mendeleev's table was periodic, or repeating, because the characteristics of elements followed a pattern. It was the forerunner to the modern periodic table.

Reading the table

The periodic table is made up of a series of groups (columns that run from top to bottom), and periods (rows that run from left to right).

Groups

Elements in the same group have the same number of electrons in their outermost shell. As you move down a group, the atoms of the elements get larger and heavier. This is because there is an increasing number of protons in the nucleus and more electrons in shells around the nucleus.



Hydrogen
An atom of hydrogen has one electron in total.

Lithium
A lithium atom has three electrons in total, with one in its outer shell.

Sodium
A sodium atom has eleven electrons in total, with one in its outer shell.

Groups
A vertical column of chemically similar elements is known as a group.

Periods
A horizontal row of chemically different elements is known as a period.

KEY

- Hydrogen
- Alkali Metals
- Alkaline Earth Metals
- Transition Metals
- Lanthanides
- Actinides
- The Boron Group
- The Carbon Group
- The Nitrogen Group
- The Oxygen Group
- Halogens
- Noble Gases

1	2	3	4	5	6	7	8	9	10	11	12
1 H Hydrogen 1.0079	2 He Helium 4.0026										
3 Li Lithium 6.941	4 Be Beryllium 9.0122										
5 Na Sodium 22.990	6 Mg Magnesium 24.305										
7 K Potassium 39.098	8 Ca Calcium 40.078	9 Sc Scandium 44.956	10 Ti Titanium 47.867	11 V Vanadium 50.942	12 Cr Chromium 51.996	13 Mn Manganese 54.938	14 Fe Iron 55.845	15 Co Cobalt 58.933	16 Ni Nickel 58.693	17 Cu Copper 63.546	18 Zn Zinc 65.39
19 Rb Rubidium 85.468	20 Sr Strontium 87.62	21 Y Yttrium 88.906	22 Zr Zirconium 91.224	23 Nb Niobium 92.906	24 Mo Molybdenum 95.94	25 Tc Technetium (96)	26 Ru Ruthenium 101.07	27 Rh Rhodium 102.91	28 Pd Palladium 106.42	29 Ag Silver 107.87	30 Cd Cadmium 112.41
31 Cs Cesium 132.91	32 Ba Barium 137.33	33 La-Lu Lanthanides 57-71	34 Hf Hafnium 178.49	35 Ta Tantalum 180.95	36 W Tungsten 183.84	37 Re Rhenium 186.21	38 Os Osmium 190.23	39 Ir Iridium 192.22	40 Pt Platinum 195.08	41 Au Gold 196.97	42 Hg Mercury 200.59
39 Fr Francium (223)	40 Ra Radium (226)	41 Ac-Lr Actinides 89-103	42 Rf Rutherfordium (261)	43 Db Dubnium (262)	44 Sg Seaborgium (266)	45 Bh Bohrium (264)	46 Hs Hassium (277)	47 Mt Meitnerium (268)	48 Ds Darmstadtium (281)	49 Rg Roentgenium (282)	50 Cn Copernicium (285)
			51 La Lanthanum 138.91	52 Ce Cerium 140.12	53 Pr Praseodymium 140.91	54 Nd Neodymium 144.24	55 Pm Promethium (145)	56 Sm Samarium 150.36	57 Eu Europium 151.96	58 Gd Gadolinium 157.25	59 Tb Terbium 158.93
			61 Ac Actinium (227)	62 Th Thorium 232.04	63 Pa Protactinium 231.04	64 U Uranium 238.03	65 Np Neptunium 237	66 Pu Plutonium 244	67 Am Americium 243	68 Cm Curium 247	69 Bk Berkelium 247

Lanthanides and actinides
The "f" block elements are sometimes called the inner transition metals.

Groups (columns)

Periods (rows)